

Final Project Guidance

Social Network Analysis, Spring 2026

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1 Background

The final project requires each group to build an **AI Agent system** for social network analysis. Each group member develops one **Skill** (a functional module), and the group assembles all Skills into a unified **Agent**.

What is a Skill? A **Skill** is a self-contained, modular unit of functionality that performs a specific task in social network analysis. Each Skill takes well-defined inputs, executes a particular function (e.g., retrieving user profiles, detecting communities, predicting links), and produces structured outputs. Skills are designed to be independently testable and reusable. In this project, each group member is responsible for developing one Skill.

What is an Agent? An **Agent** is a system that orchestrates multiple Skills to accomplish a complex, high-level task. The Agent defines the overall workflow: it decides which Skills to invoke, in what order, and how to combine their outputs to produce a final result. For example, an Agent for social influence analysis might coordinate a data collection Skill, a network construction Skill, a centrality computation Skill, and a visualization Skill into a coherent pipeline. In this project, each group builds one Agent by integrating all members' Skills.

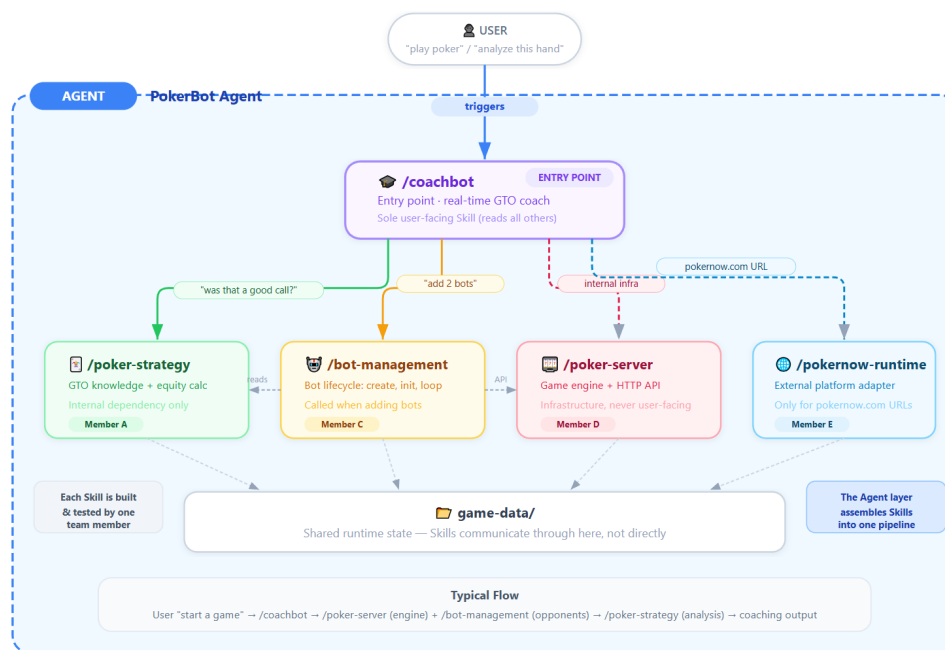


Figure 1: Architecture of the PokerBot Agent. The Agent layer assembles Skills into a coherent pipeline, and Skills communicate through a shared data layer.

A concrete example: PokerBot in StudyClawHub (see Fig. 1). PokerBot is an AI Agent that plays Texas Hold'em with real-time coaching. It is composed of five Skills, each developed independently: a poker-strategy Skill (GTO knowledge base and equity calculators), a coachbot Skill (real-time coaching personality), a bot-management Skill (AI opponent lifecycle), a poker-server Skill (game engine and state API), and a pokernow-runtime Skill (adapter for an external poker platform). No Skill depends on the internal implementation of another; they communicate only through a shared data layer. The Agent orchestrator decides when to invoke each Skill based on user input (e.g., “start a game” triggers poker-server and bot-management; asking “was that a good call?” triggers poker-strategy and coachbot). Your social network analysis Agent should follow the same pattern.

Learning Objectives. Through this project, students are expected to:

1. Learn to use AI-powered development platforms including **OpenClaw**, **Claude Code**, and **WorkBuddy**.
2. Understand the concepts of **Agents** and **Skills**, and apply them to solve real-world social network analysis tasks.
3. Practice the open-source workflow in the AI era, including collaborative development, code publishing, and technical report writing.

2 StudyClawHub

All Skills and Agents must be published to **StudyClawHub**, a lightweight skill registry for social network mining powered by GitHub. StudyClawHub allows students to browse, install, and share Skills and Agents with zero server infrastructure.

Submission Steps.

1. **Submit via tools.** Each member submits their individual Skill, and the group submits the Agent that integrates all component Skills. Submission can be done through **Claude Code**, **OpenClaw**, or **WorkBuddy**.
2. **Register on the website.** Go to the StudyClawHub website and click “**Submit Skill**” or “**Submit Agent**” to fill in the metadata (name, description, version, tags, GitHub repo URL, and author) and complete the registration.

3 Project Requirements and Report Format

Deliverables and Deadline. Each group must submit the following by **May 15, 2026**. Presentations will take place during **Week 14**.

1. **Group report (Agent):** 4 pages, NeurIPS format.
2. **Individual report (Skill):** 3 pages per member, NeurIPS format.
3. **Code submission:** Agent and all Skills published to StudyClawHub.
4. **Presentation:** each group presents their Agent in Week 14.

Group Formation. Each group consists of **at most 5 members**. Each member must contribute exactly **one Skill** to the group Agent.

Agent Design (Group). The group should design and implement an Agent that performs a meaningful task related to social network analysis. The Agent should: *(i)* integrate all members’ Skills into a coherent system; *(ii)* demonstrate a clear workflow where Skills interact or compose to achieve the Agent’s overall goal; *(iii)* be evaluated on a well-defined task with appropriate metrics.

Skill Development (Individual). Each member develops one Skill that serves as a functional module within the Agent. A Skill should: *(i)* have a clearly defined function (e.g., community detection, influence prediction, link prediction, sentiment analysis, information retrieval, or visualization); *(ii)* be individually testable and produce measurable results; *(iii)* include references to relevant prior work or methods.

Table 1: Report specifications. References do not count toward the page limit.

Report Type	Page Limit	Scope
Group Report (Agent)	4 pages	Overall Agent design, integration, and evaluation
Individual Report (Skill)	3 pages	Individual Skill method, implementation, and results

Report Format. Both reports must use the **NeurIPS format** (this template) and include the following five components. We have published a **LatexReport Skill** on StudyClawHub that can automatically configure the NeurIPS format for you.

- **Functionality.** Clearly describe what the Agent/Skill does, including input/output specifications and how it addresses the target task.
- **References.** Cite relevant prior work, methods, or tools that your design builds upon. Discuss how your approach relates to existing literature.
- **Results.** Present quantitative or qualitative results with appropriate evaluation metrics.
- **Analysis.** Analyze the results: what works, what does not, and why. Ablation studies or error analysis are encouraged.
- **Visualization.** Provide at least one meaningful visualization (e.g., network graphs, performance plots, workflow diagrams).

4 Suggested Topics

Below are example project topics. You are encouraged to propose your own topic as long as it involves building an Agent with multiple Skills.

1. **Social Network Analysis Agent.** Build an Agent that automates end-to-end analysis of a social network. Given a raw dataset or a platform URL, the Agent should be able to crawl or parse the data, construct a graph representation, run community detection and influence ranking algorithms, and produce a final report with network statistics and visualizations. The user should be able to specify analysis goals in natural language (e.g., “find the top influencers in this network” or “show me the community structure”), and the Agent should select and execute the appropriate analysis pipeline automatically. The Agent should support at least two different community detection methods and allow the user to compare their results.
2. **Daily arXiv Research Briefing Agent.** An Agent that monitors arXiv on a daily basis and generates a personalized research briefing. The user provides a research topic or a set of keywords, and the Agent fetches new papers, ranks them by relevance, extracts key contributions and methods from each paper, and compiles a structured daily report. The report should include a summary table, highlight papers with the highest novelty or relevance, and support follow-up queries such as “show me papers related to graph neural networks from this week.”
3. **Homework Feedback Agent.** An Agent that automates the grading and feedback process for course assignments. Given a batch of student submissions and a reference solution or rubric, the Agent should evaluate each submission for correctness, identify common error patterns across the class, generate individualized feedback with suggestions for improvement, and produce a class-level analytics dashboard showing score distributions and frequent mistakes. The feedback should be constructive and reference specific parts of the student’s work.
4. **LLM Hallucination Detection via Cross-Verification.** An Agent that verifies the factual accuracy of LLM-generated text. Given a passage produced by an LLM, the Agent should extract key factual claims, retrieve relevant evidence from web sources or databases, and assess each claim’s veracity. The final output should be an annotated version of the original text where each claim is labeled as supported, unsupported, or contradicted, along with links to the evidence sources and an overall reliability score.

5. **Misinformation Detection Agent.** An Agent that monitors social media platforms and detects the spread of misinformation. The Agent should collect posts and their propagation networks, analyze both content features (e.g., sentiment, linguistic cues) and structural features (e.g., propagation speed, network topology of sharers), assign credibility scores to sources, and generate alerts when potential misinformation is detected. The system should be able to track how a specific claim evolves and spreads across the network over time.
6. **Academic Collaboration Recommender.** An Agent that helps researchers find potential collaborators. Given a researcher’s name or publication list, the Agent should construct a co-authorship network from publication databases (e.g., DBLP, Semantic Scholar), identify research communities using graph clustering algorithms, compute topic similarity between researchers, and recommend collaborators ranked by a combination of topical alignment and network proximity. The Agent should also be able to visualize the researcher’s ego network and explain why each recommendation is made.
7. **Restaurant/Travel Recommendation Agent.** An Agent that provides personalized dining or travel recommendations by mining social platform data. The Agent should scrape and aggregate user reviews from platforms such as Dianping, Meituan, or Xiaohongshu, build a user preference profile from review history or stated preferences, perform sentiment analysis on reviews, and generate ranked recommendations with explanations. For travel recommendations, the Agent should additionally plan an itinerary that accounts for geographic proximity and time constraints.
8. **Job Market Intelligence Agent.** An Agent that analyzes the job market by mining recruitment platforms and professional networks. The Agent should collect job postings and professional profiles, extract and normalize skill requirements, build a skill co-occurrence network to identify skill clusters, and track demand trends over time. For a student user, the Agent should compare the student’s current skill set against market demands, identify gaps, and recommend specific positions. The system should support queries such as “what skills are trending for data science roles in 2026?”
9. **Self-Evolving Campus Q&A Agent.** An Agent that answers questions about campus life and continuously improves itself based on user feedback. The Agent should handle queries about course selection, campus facilities, club activities, administrative procedures, and daily logistics. When a user marks an answer as incorrect or unhelpful, the Agent should retrieve better information from campus websites, forums, or knowledge bases, and update its internal knowledge and prompt strategies accordingly. Over multiple interaction rounds, the Agent should demonstrably improve its answer accuracy and coverage on recurring question types without manual intervention.
10. **Campus Second-Hand Trading Agent.** An Agent that facilitates second-hand transactions among students on campus. The Agent should collect and parse item listings from campus forums or chat groups, build a transaction network among buyers and sellers, match buyers to relevant listings based on stated preferences and budget, and detect potentially fraudulent or suspicious postings through anomaly analysis on the transaction graph. The Agent should also estimate fair market prices for listed items by analyzing historical transaction data and similar listings. The system should support natural language queries such as “find me a used iPad under 2000 yuan” and provide trust scores for sellers based on their transaction history.
11. **Gaming Strategy Coach Agent.** An Agent that serves as a personal coach for multiplayer games. Given match history data (e.g., from League of Legends, Honor of Kings, or similar games), the Agent should construct player interaction networks, analyze team compositions and win/loss patterns, identify the user’s strengths and weaknesses through statistical profiling, and generate actionable strategy suggestions tailored to the user’s play style and rank. The Agent should support queries such as “why did I lose this match?” and provide analysis grounded in specific match data.